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Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements is correct?

A. We can model a hummingbird as a perfect sphere when analyzing the detailed motion of its wings.
This is incorrect because wing motion depends strongly on shape and rotation, so a sphere model removes the key physics.

B. We can reasonably treat a train as a point mass when estimating how its position changes over a 50 km trip.
This is correct because over large travel distances the train's size is negligible compared with the displacement, so a point-mass model for position is appropriate.

C. We should never assume motion occurs in a vacuum, even if we explicitly decide to neglect air drag.
This is incorrect because "vacuum" is often an idealization used precisely to ignore air resistance when it is negligible.

D. None of the above is correct.
This is incorrect because option B gives a valid scale-based approximation.

Q2. Which of the following equations is correct?

A. 1 mm = 1 m.
This is incorrect because a millimeter is much smaller than a meter.

B. 1 = 0.001.
This is incorrect because it equates a dimensionless 1 to a different number without units or context.

C. 1 mm = 0.001 m.
This is correct because the prefix milli- means 10^{-3}, so 1 mm = 10^{-3} m.

D. 1000 mm = 0.001 m.
This is incorrect because 1000 mm equals 1 m, not 0.001 m.

Q3. Which option has the incorrect unit?

A. Pressure: pascal (N/m^2).
This is correct because pressure is force per area, with SI unit pascal.

B. Momentum: $kg \cdot m/s$.
This is correct because momentum has units of mass times velocity ($kg \cdot m/s$).

C. Electric current: coulomb.
This is incorrect because current is charge per time with unit ampere (C/s), not coulomb.

D. Work: joule.
This is correct because work/energy is measured in joules.

Q4. A rescue drone is launched from a base camp located at the origin and detects a flare at the position $\vec{r} = 4\hat{\imath} + 7\hat{\jmath}$, where distances are measured in hectometers. Which of the following statements is correct?

A. The distance between the flare and the base camp is 11 hectometers.
This is incorrect because 11 hm comes from adding components $(4 + 7)$ instead of using the Pythagorean distance.

B. The distance between the flare and the base camp is 7 hectometers.
This is incorrect because it confuses the y-component with the total distance from the origin.

C. The distance between the flare and the base camp is 4 hectometers.
This is incorrect because it confuses the x-component with the total distance from the origin.

D. None of the above is correct.
This is correct because the distance is $\sqrt{4^2 + 7^2} = \sqrt{65} \approx 8.1$ hm, which is not 11, 7, or 4.

Q5. If the displacement of an object is given by the function $\vec{s} = ((0.80\ \text{m/s}^4)t^4 + (1.50\ \text{m/s}^3)t^3)\,\hat{\imath}$, which of the following statement is correct?

A. This is a constant-acceleration motion.
This is incorrect because differentiating twice yields an acceleration that depends on t, not a constant.

B. The magnitude of the acceleration is 1.50 m/s².
This is incorrect because 1.50 is a coefficient in the displacement term and does not equal a constant acceleration.

C. The magnitude of the acceleration at $t = 0$ is 0 m/s².
This is correct because $a(t) = \frac{d^2s}{dt^2} = (9.6t^2 + 9.0t)\hat{\imath}$, giving $a(0) = 0$.

D. The magnitude of the acceleration at $t = 0$ is 9.0 m/s².
This is incorrect because the $9.0t$ term vanishes at $t = 0$, so the acceleration is not 9.0 m/s^2 then.

Q6. A robot takes 4 s to move from position A ($\vec{r}_A = 1.0\hat{\imath} - 2.0\hat{\jmath}$) to position B ($\vec{r}_B = 7.0\hat{\imath} + 6.0\hat{\jmath}$) along a straight path, where length units are in meters. Which of the following statements is correct?

A. The moving distance is 10 m.
This is correct because the straight-line distance is $|\vec{r}_B - \vec{r}_A| = \sqrt{6^2 + 8^2} = 10$ m.

B. The speed at $t = 0$ must be 2.5 m/s.
This is incorrect because knowing only endpoints and total time does not determine the instantaneous speed at $t = 0$.

C. The acceleration at $t = 0$ must be 0.63 m/s².
This is incorrect because endpoint data alone does not uniquely fix the acceleration at a specific time.

D. None of the above is correct.
This is incorrect because the distance statement in option A is determinable and true.

Q7. A tennis ball is launched with speed u_0 at an angle ϕ above the horizontal. What is the speed of the ball when it is at its maximum height (neglect air resistance)?

A. u_0.
This is incorrect because the vertical component decreases to zero at the top, reducing the speed from its initial value.

B. $u_0 \cos \phi$.
This is correct because at maximum height the vertical velocity is zero while the horizontal component remains $u_0 \cos \phi$.

C. $u_0 \sin \phi$.
This is incorrect because $u_0 \sin \phi$ is the initial vertical component, which is not the speed at the top.

D. 0.
This is incorrect because the horizontal component is generally nonzero at the top, so the speed is not zero.

Q8. A rover initially moves with velocity $\vec{v}_0 = (12.0\ \text{m/s})\hat{\imath}$ when a guidance system applies a constant acceleration $\vec{a} = (-1.50\ \text{m/s}^2)\hat{\imath} + (0.80\ \text{m/s}^2)\hat{\jmath}$. Based on this scenario, which of the following statements is correct?

A. The x-component of the rover's velocity remains constant.
This is incorrect because a nonzero a_x changes v_x over time.

B. The y-component of the rover's velocity remains constant.
This is incorrect because a nonzero a_y makes the y-velocity change from its initial value (here, from 0 upward/downward).

C. Both the direction of motion and speed change.
This is correct because the acceleration adds a y-velocity component (changing direction) and also changes v_x (changing speed).

D. None of the above is correct.
This is incorrect because option C correctly describes the effect of having both a_x and a_y nonzero.

Q9. If the magnitudes of both charges are multiplied by 5 and their separation is multiplied by 2, what happens to the magnitude of the electrostatic force that each charge exerts on the other?

A. It stays the same.
This is incorrect because scaling q_1 and q_2 changes the numerator in Coulomb's law.

B. It decreases by a factor of 4.
This is incorrect because increasing distance by 2 makes the force smaller by 4, but the charges also increase and overwhelm that reduction.

C. It increases by a factor of 5.
This is incorrect because both charges change, giving a factor of 25 in the numerator, not 5.

D. It increases by a factor of 25.
This is incorrect because the distance also changes, contributing a factor of 4 in the denominator.

E. It increases by a factor of 25/4.
This is correct because $F \propto q_1 q_2 / r^2$, so the factor is $(5 \cdot 5)/(2^2) = 25/4$.

Q10. An object can escape a planet with radius R and mass M at an escape speed given by $v = \sqrt{2GM/R}$, where G is the universal gravitational constant. Based on this expression for escape speed, which of the following statements is correct?

A. Doubling the mass of the escaping object doubles the escape speed.
This is incorrect because the formula contains only the planet's M and R, not the object's mass.

B. Keeping M fixed but making R larger increases the escape speed.
This is incorrect because $v \propto 1/\sqrt{R}$, so a larger radius decreases escape speed if mass is unchanged.

C. A planet with smaller mass and smaller radius always has the largest escape speed.
This is incorrect because escape speed increases with larger M and decreases with larger R, so you must compare both together rather than assume.

D. A planet with greater mass and smaller radius requires a higher escape speed.
This is correct because $v = \sqrt{2GM/R}$ grows with increasing M and decreasing R.

Q11. The force $\vec{F} = (5.20\text{ N})\hat{\imath} + (-1.70\text{ N})\hat{\jmath}$ acts on a particle as it moves along the x axis from $x = -1.00$ m to $x = 2.50$ m. What is the work, in joules, done by the force on the particle?

A. 6.0 J.
This is incorrect because it effectively uses the wrong component or an incorrect displacement magnitude.

B. 18.2 J.
This is correct because $\Delta\vec{r} = (3.50\text{ m})\hat{\imath}$ so $W = \vec{F} \cdot \Delta\vec{r} = F_x \Delta x = (5.20)(3.50) = 18.2$ J.

C. −6.0 J.
This is incorrect because motion is purely in x, so the negative F_y does not contribute to work here.

D. 24.0 J.
This is incorrect because it corresponds to using the full force magnitude instead of the dot product with the displacement.

Q12. A 3.0 kg cart moving at 4.0 m/s hits a bumper and rebounds at 2.0 m/s in the opposite direction. The collision lasts 0.15 s. What is the average force exerted on the cart during the collision?

A. 80 N.
This is incorrect because it underestimates the impulse by not accounting for the full velocity change including the reversal.

B. 120 N.
This is correct because taking forward as positive gives $\Delta v = -2.0 - 4.0 = -6.0$ m/s so $|F_{\text{avg}}| = m|\Delta v|/\Delta t = 3.0 \cdot 6.0/0.15 = 120$ N.

C. 40 N.
This is incorrect because it effectively uses only part of the velocity change or an incorrect collision time.

D. None of the above is correct.
This is incorrect because 120 N is a valid value from the impulse-momentum calculation.

Q13. A 65 kg outfielder is initially at rest and catches a 145 g baseball moving horizontally at 90 km/hr. Assuming the ball sticks in the glove, what is the outfielder's speed immediately after the catch?

A. 0.056 m/s.
This is correct because momentum conservation gives $v_f = \frac{(0.145)(90/3.6)}{65+0.145} \approx 0.056$ m/s.

B. 0.045 m/s.
This is incorrect because it typically comes from using the wrong total mass (forgetting to add the ball) or rounding the speed conversion incorrectly.

C. 5.6 m/s.
This is incorrect because it misplaces decimals, effectively treating grams as kilograms or failing to convert km/hr to m/s.

D. 0.36 m/s.
This is incorrect because it overestimates the result by not using the large combined mass in the denominator.
